

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of seed mass per plot (esy4)

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(DHARMA)
library(emmeans)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.seedmass)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 0.001574195 0.001489662 0.001405129
```

```
median(sites$y)
```

```
## [1] 0.00122
```

```
sd(sites$y)
```

```
## [1] 0.001072695
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
```

```
## 0.00038 0.00347
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct> <int>
## 1 654      203
## 2 664      203
## 3 686      215
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord> <int>
## 1 positive    102
## 2 restored    401
## 3 negative    118
```

```
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R      330
## 2 R22    210
## 3 R1A     81
```

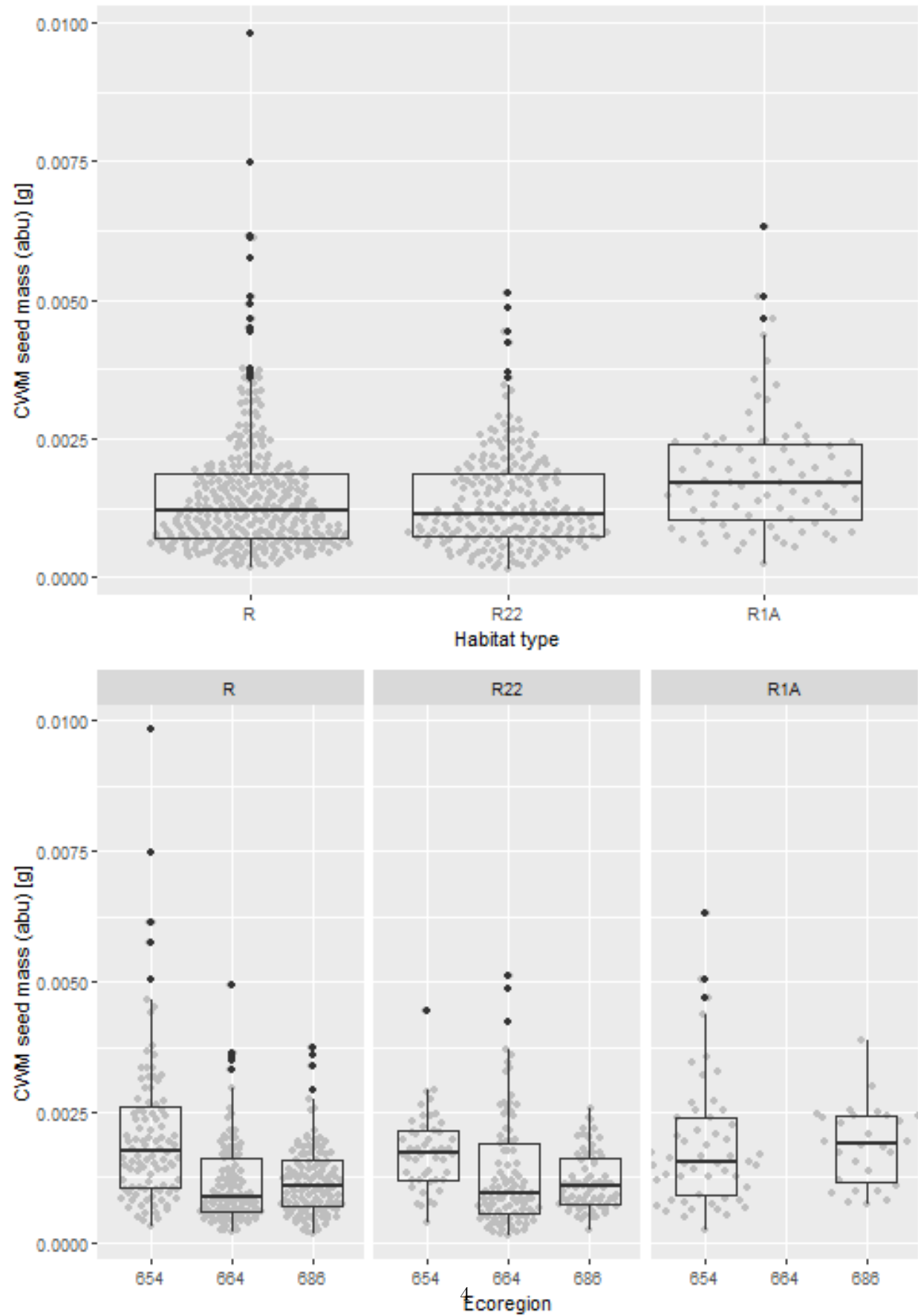
```
sites %>% count(esy4, eco.id)
```

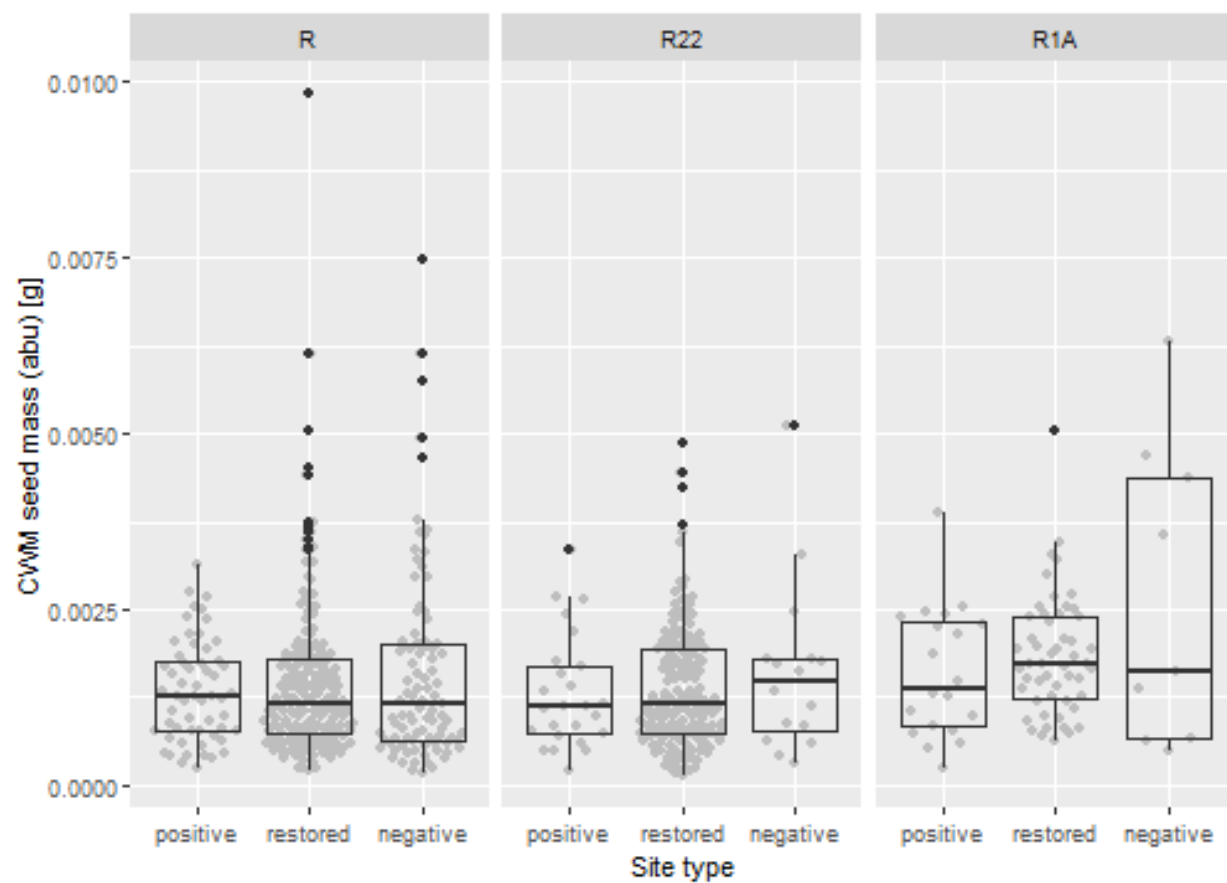
```
## # A tibble: 8 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R      654     102
## 2 R      664     112
## 3 R      686     116
## 4 R22    654      48
## 5 R22    664      91
## 6 R22    686      71
## 7 R1A    654      53
## 8 R1A    686      28
```

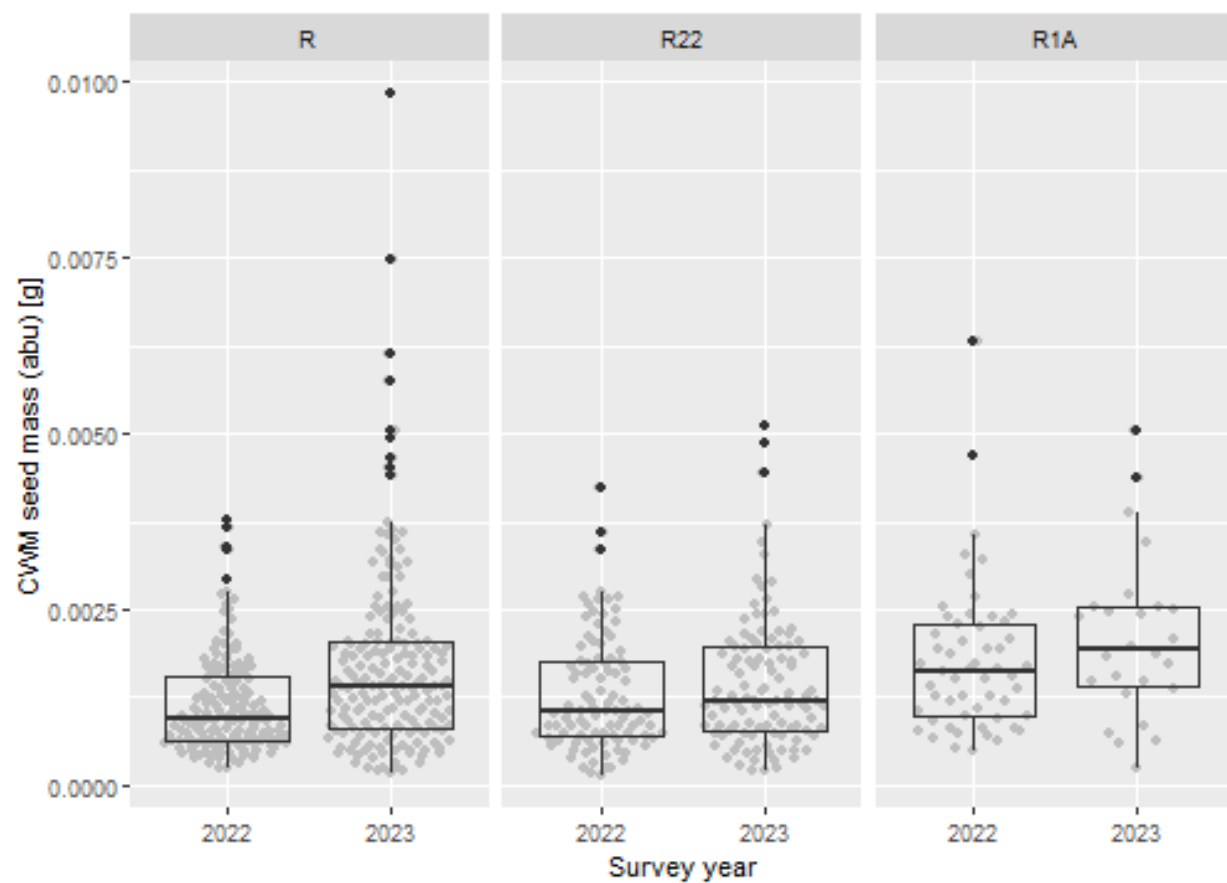
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 9 x 3
##   esy4 site.type      n
##   <fct> <ord> <int>
## 1 R      positive     57
## 2 R      restored    180
## 3 R      negative     93
## 4 R22    positive     25
## 5 R22    restored    169
## 6 R22    negative     16
## 7 R1A    positive     20
## 8 R1A    restored     52
## 9 R1A    negative      9
```

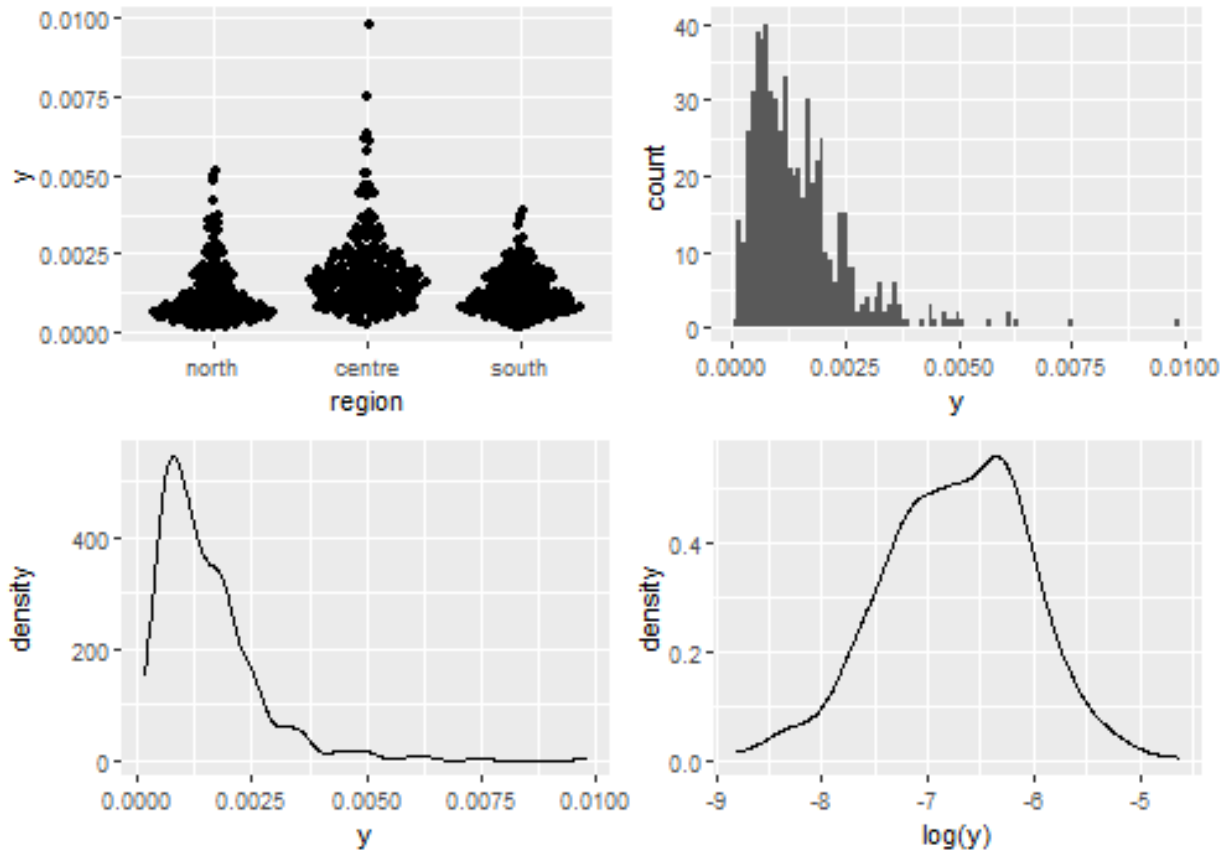
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum.)) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_seedmass_esy4_1.Rdata"))
load(file = here("outputs", "models", "model_seedmass_esy4_3.Rdata"))
m_1 <- m1
m_2 <- m3

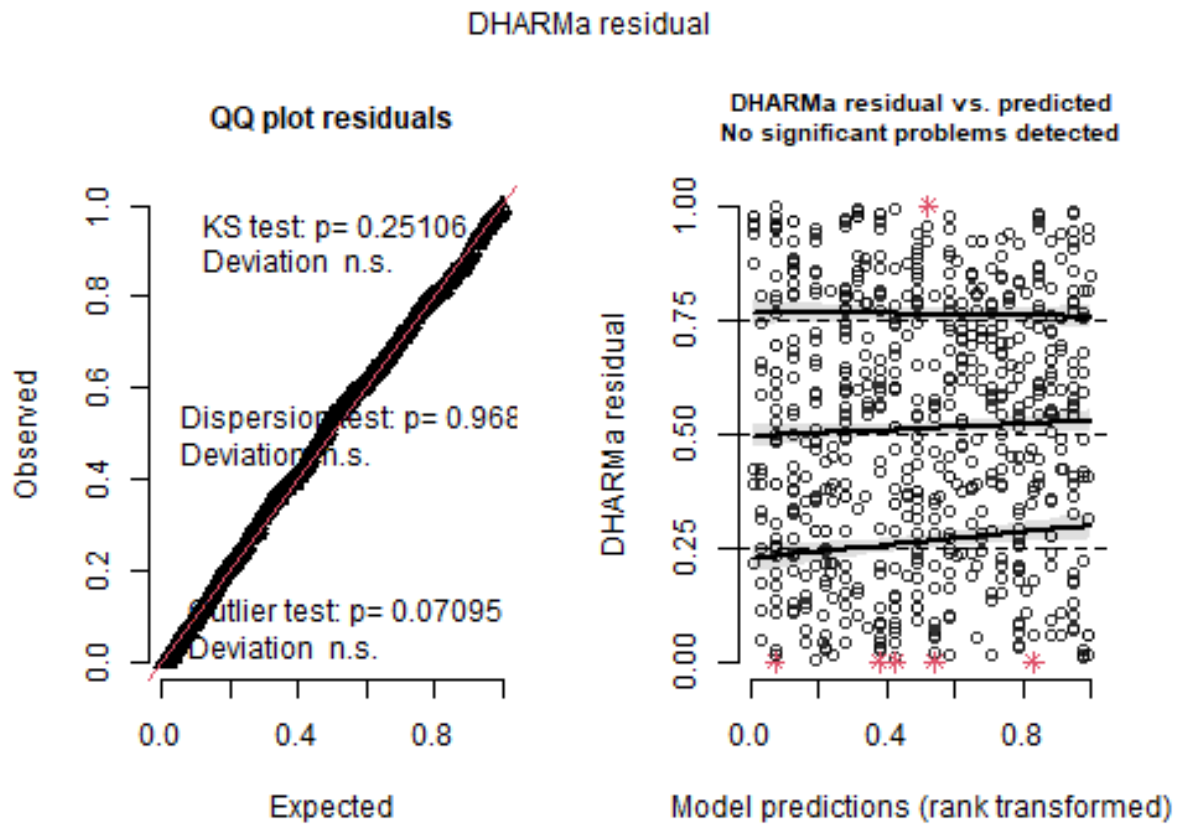
m_1@call
## lmer(formula = log(y) ~ esy4 * (site.type + eco.id) + obs.year +
##       (1 | id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = log(y) ~ esy4 * (site.type + eco.id) + obs.year +
##      hydrology + (1 | id.site), data = sites, REML = FALSE)
```

Model check

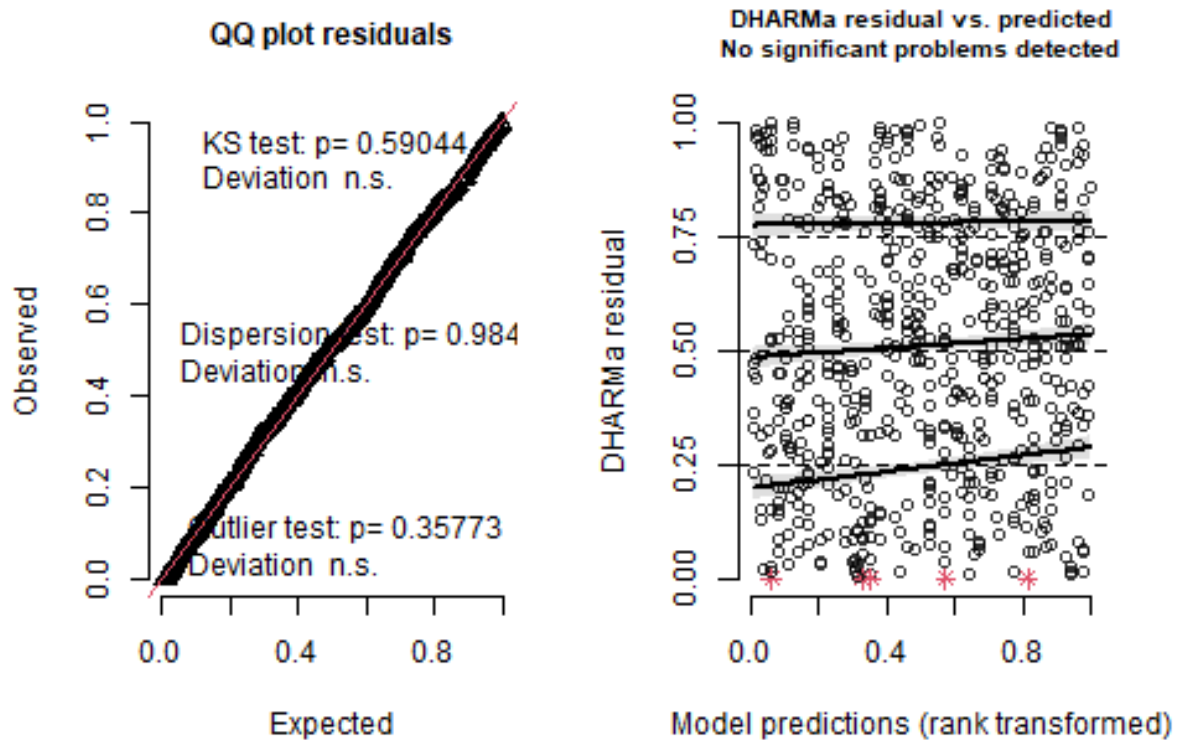
DHARMa

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

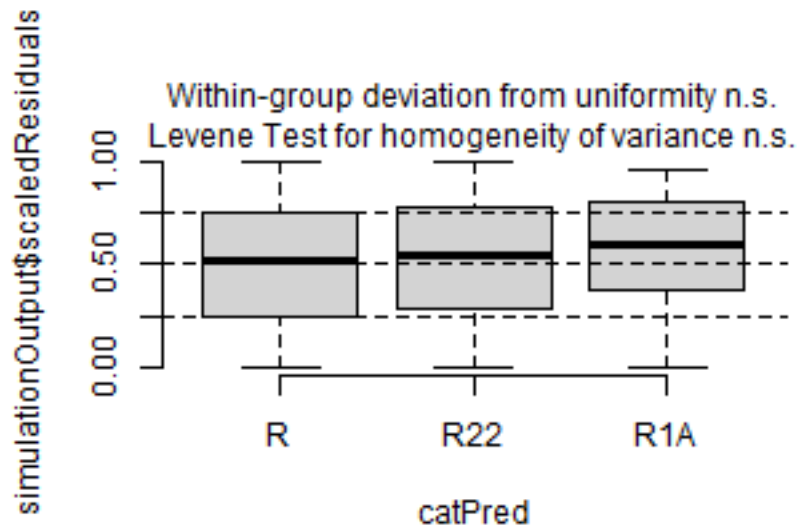


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

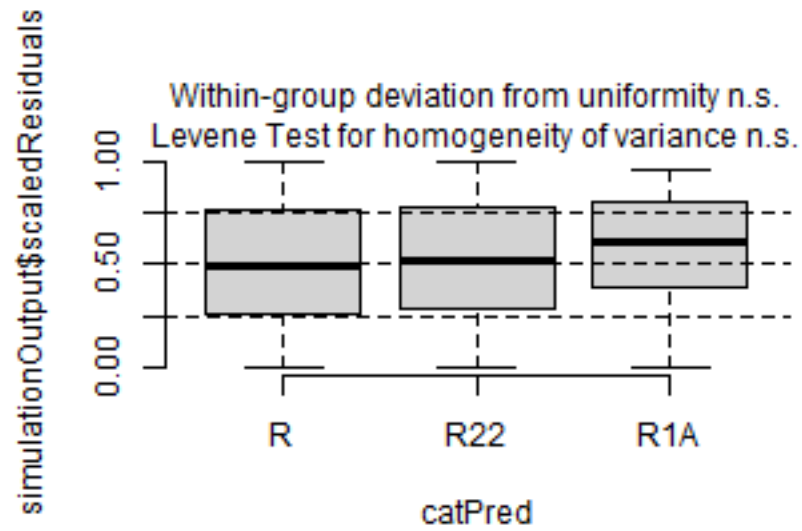

DHARMa residual



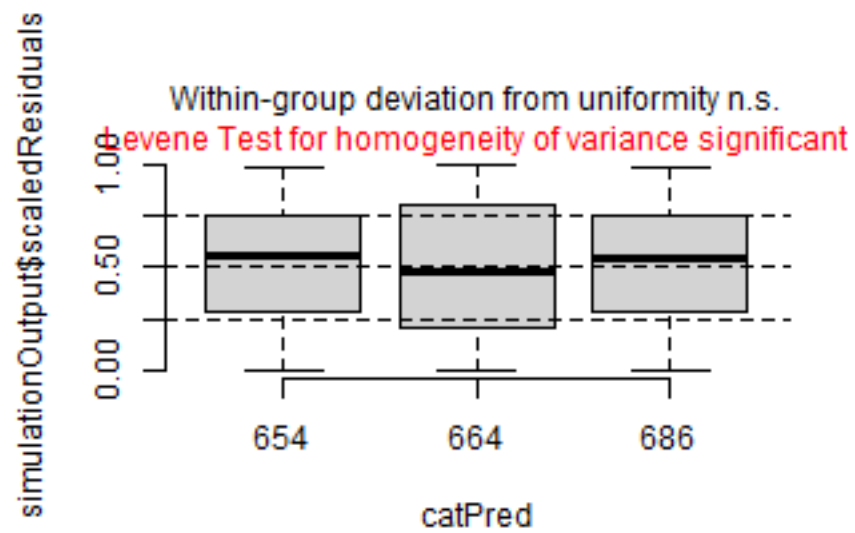
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



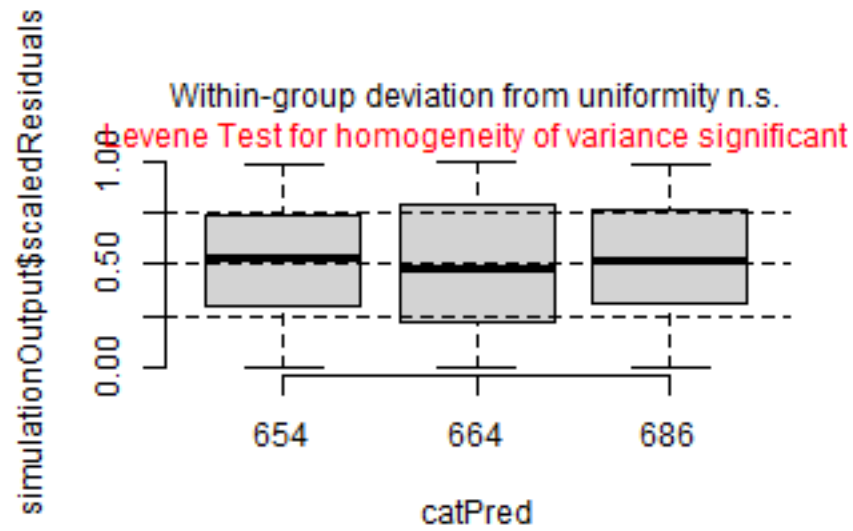
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



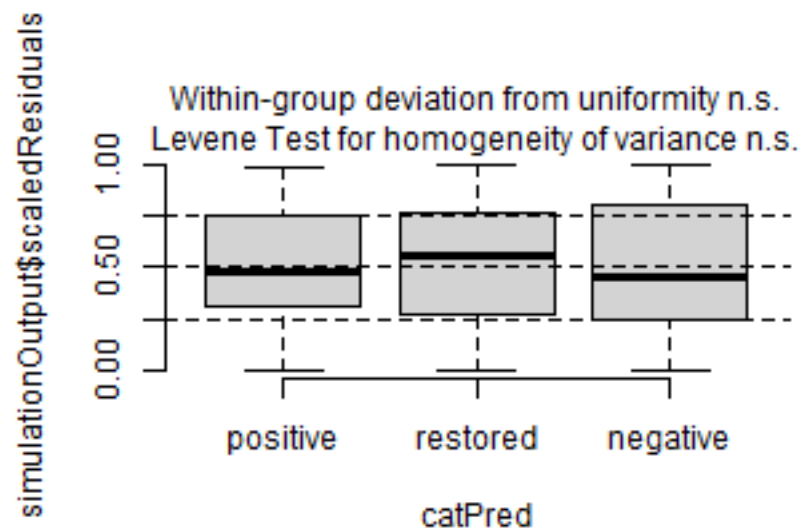
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



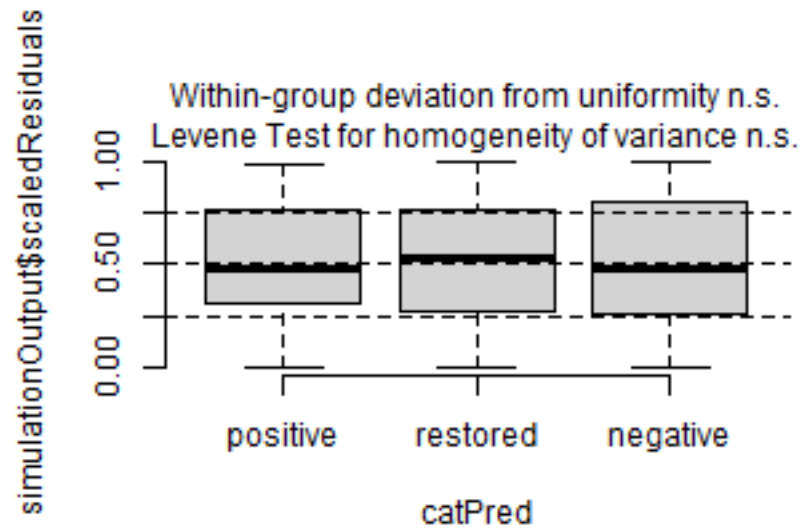
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



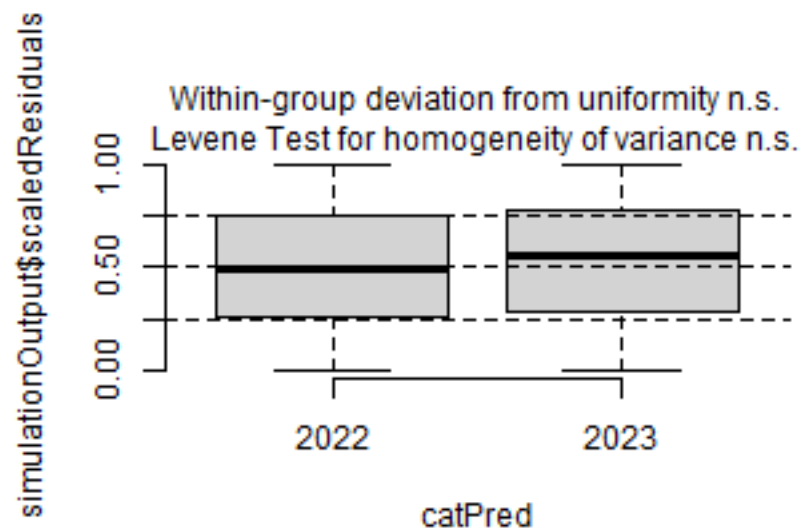
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



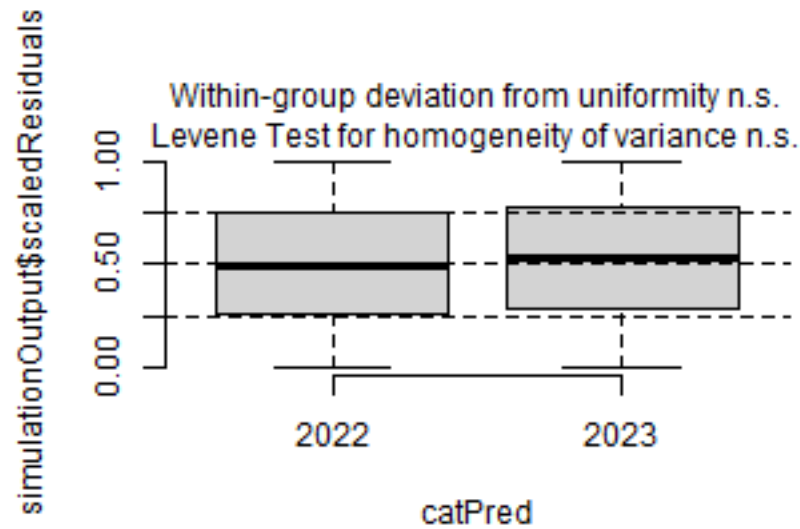
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



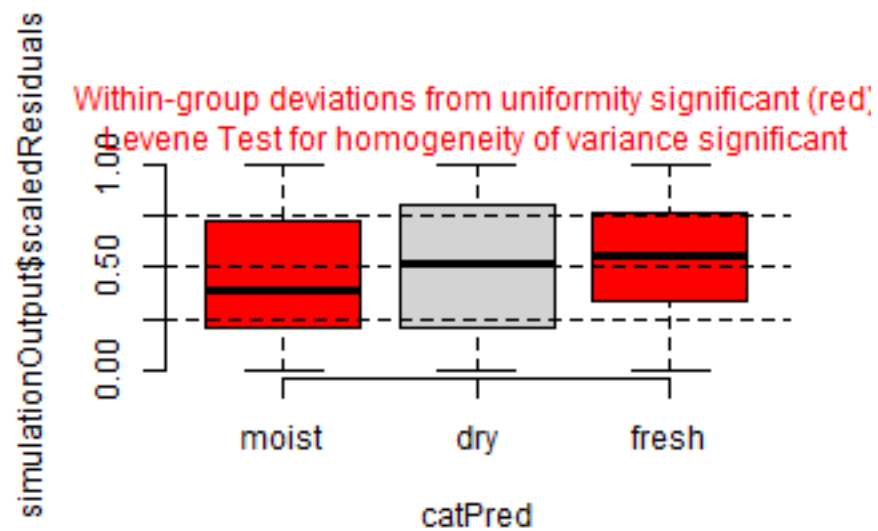
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



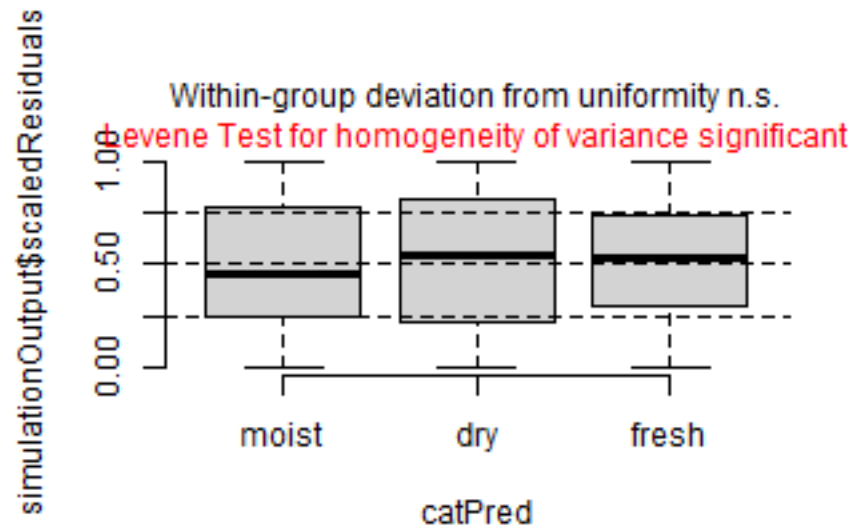
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



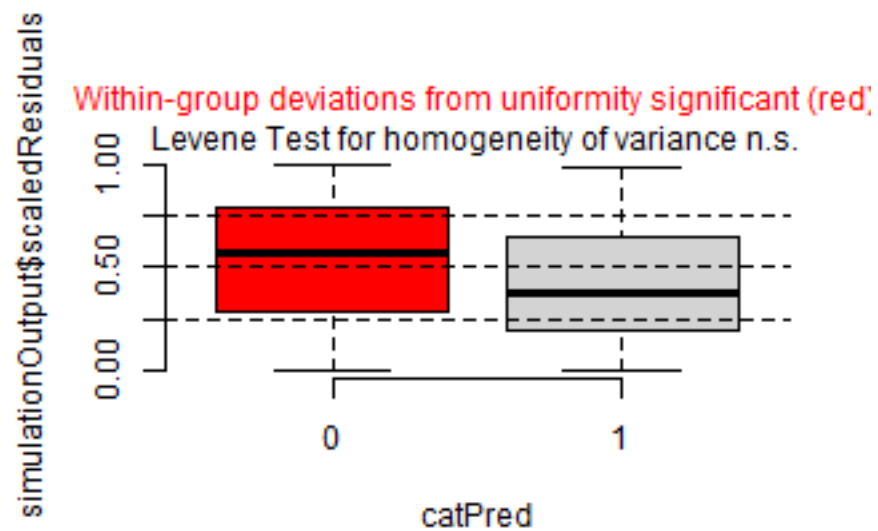
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



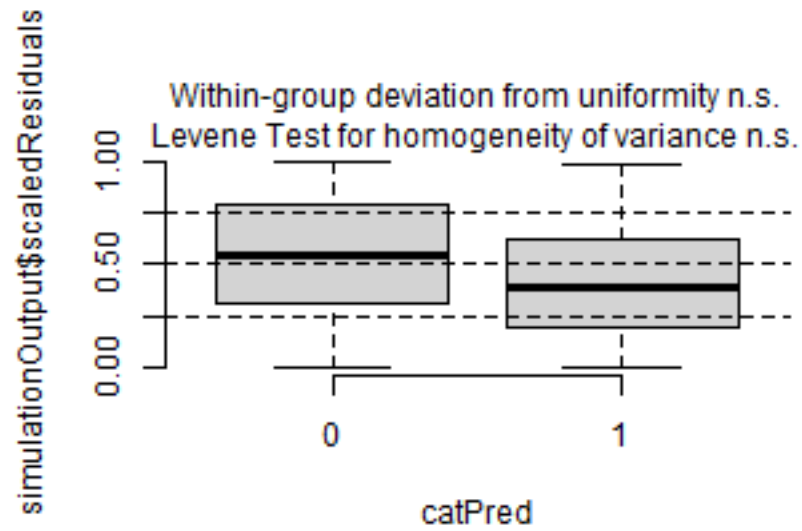
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



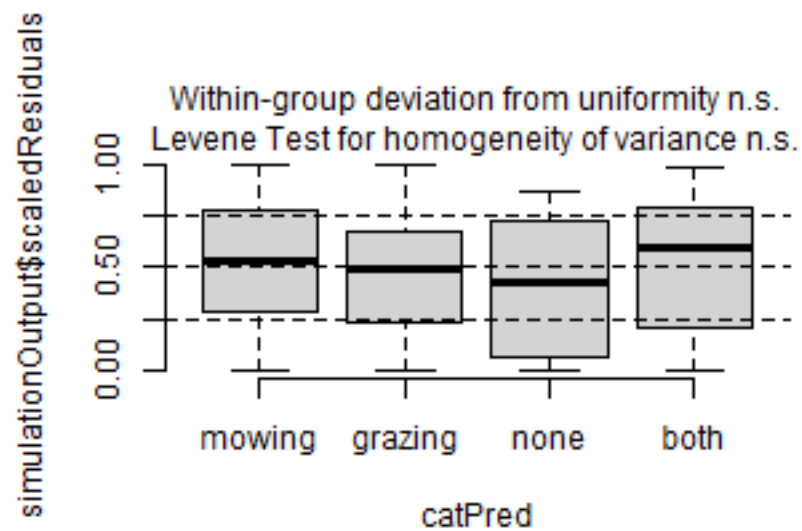
```
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



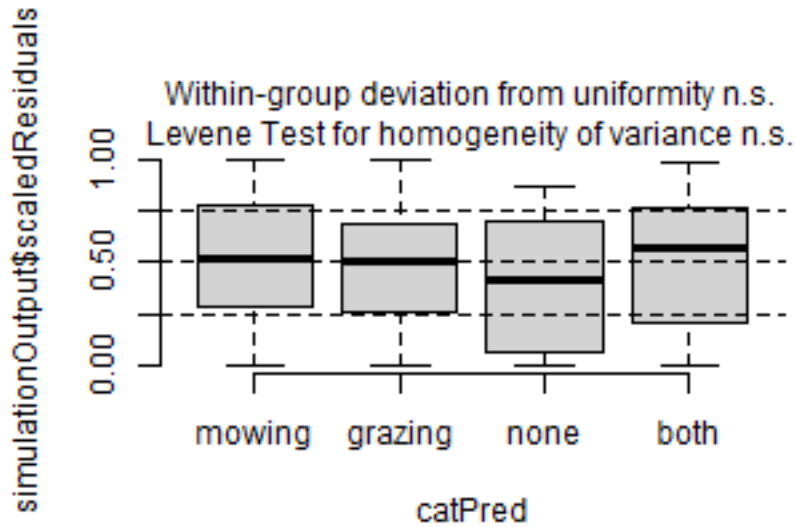
```
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove $VIF > 3$ or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##                GVIF Df  GVIF^(1/(2*Df))
## esy4           11.330679 2      1.834695
## site.type      1.418765 2      1.091384
## eco.id         1.594355 2      1.123689
## obs.year       1.022266 1      1.011072
## esy4:site.type  4.999271 4      1.222822
## esy4:eco.id     9.139388 3      1.445949
```

```
car::vif(m_2)
```

```
##                GVIF Df  GVIF^(1/(2*Df))
## esy4           12.586319 2      1.883539
## site.type      1.504139 2      1.107445
## eco.id         1.630581 2      1.130019
## obs.year       1.029409 1      1.014598
## hydrology      1.300347 2      1.067861
## esy4:site.type  5.063599 4      1.224778
## esy4:eco.id     9.354352 3      1.451562
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##          R2m          R2c
## [1,] 0.1409498 0.6805522
MuMIn::r.squaredGLMM(m_2)
##          R2m          R2c
## [1,] 0.149761 0.6790846
```


AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMin::AICc(m_1, m_2) %>%  
  arrange(AICc)  
##      df      AICc  
## m_1 17 955.5114  
## m_2 19 956.6572
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)  
  
## Analysis of Deviance Table (Type III Wald chisquare tests)  
##  
## Response: log(y)  
##  
##           Chisq Df Pr(>Chisq)  
## (Intercept)  2877.1107  1 < 2.2e-16 ***  
## esy4          0.5213  2  0.77055  
## site.type     0.7991  2  0.67062  
## eco.id        23.2769  2  8.82e-06 ***  
## obs.year      5.2836  1  0.02153 *  
## hydrology     3.1367  2  0.07389  
## esy4:site.type 3.4804  4  0.48087  
## esy4:eco.id    3.8800  3  0.27471  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
summary(m_2)  
  
## Linear mixed model fit by maximum likelihood ['lmerMod']  
## Formula: log(y) ~ esy4 * (site.type + eco.id) + obs.year + hydrology +  
## (1 | id.site)  
## Data: sites  
##  
##           AIC          BIC      logLik -2*log(L)  df.resid  
##      955.4       1039.6     -458.7      917.4      602  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -3.9928 -0.4799  0.0140  0.4601  2.8670  
##  
## Random effects:  
## Groups   Name            Variance Std.Dev.  
## id.site  (Intercept)  0.2472    0.4972  
## Residual                0.1499    0.3871  
## Number of obs: 621, groups: id.site, 177  
##  
## Fixed effects:  
##              Estimate Std. Error t value  
## (Intercept)    -6.55120    0.12214 -53.639  
## esy4R22         -0.04730    0.10199  -0.464  
## esy4R1A          0.07028    0.14104   0.498
```

```
## site.type.L      -0.09094    0.10684   -0.851
## site.type.Q      -0.01542    0.07743   -0.199
## eco.id664        -0.52268    0.11482   -4.552
## eco.id686        -0.41497    0.11180   -3.712
## obs.year2023      0.19000    0.08266    2.299
## hydrologyfresh    0.10291    0.10752    0.957
## hydrologymoist    -0.07868    0.12397   -0.635
## esy4R22:site.type.L 0.15406    0.12845    1.199
## esy4R1A:site.type.L 0.36780    0.24447    1.504
## esy4R22:site.type.Q 0.02517    0.08767    0.287
## esy4R1A:site.type.Q 0.12844    0.16818    0.764
## esy4R22:eco.id664   0.03455    0.12635    0.273
## esy4R22:eco.id686  -0.03764    0.11714   -0.321
## esy4R1A:eco.id686   0.32621    0.18321    1.781

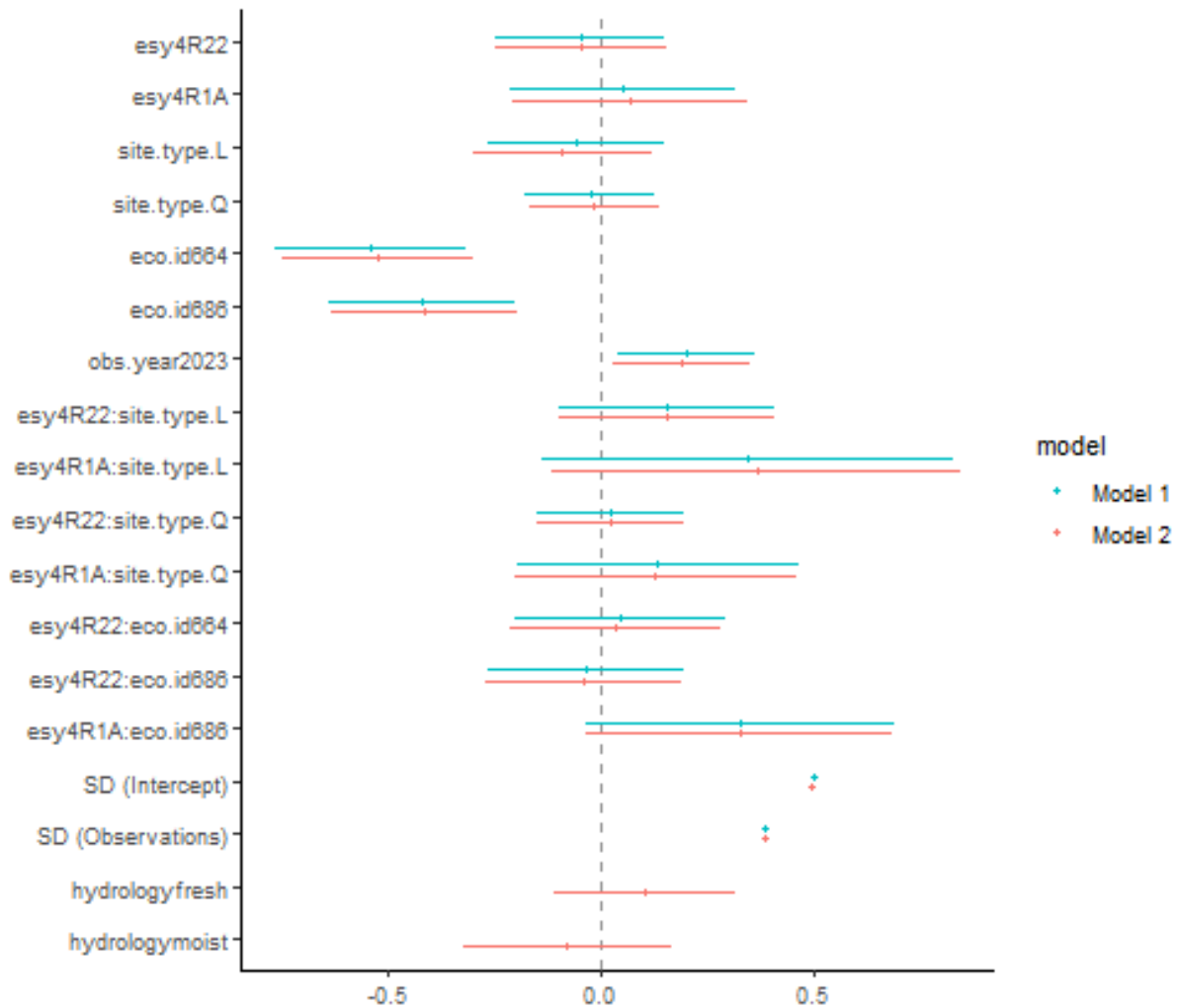
##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE) or
##     vcov(x)         if you need it

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(m_2, "esy4", type = "response"))
```

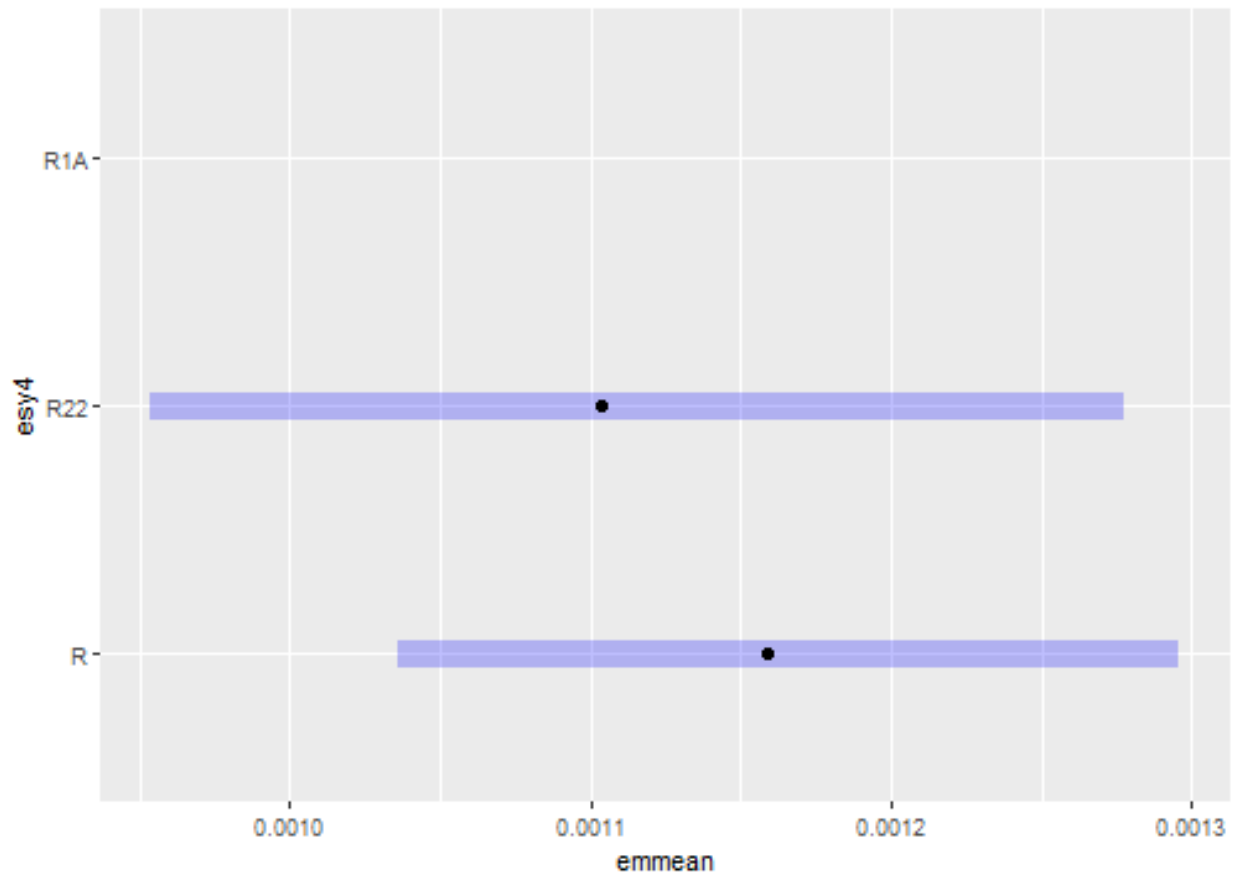
ESy EUNIS Habitat type

```
## esy4 response      SE  df lower.CL upper.CL
## R      0.00116 6.58e-05 156 0.001036 0.00130
## R22     0.00110 8.19e-05 289 0.000954 0.00128
## R1A    nonEst      NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
plot(emm, comparison = FALSE)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
(emm <- emmeans(
  m_2,
  revpairwise ~ eco.id + esy4,
  type = "response"
))
```

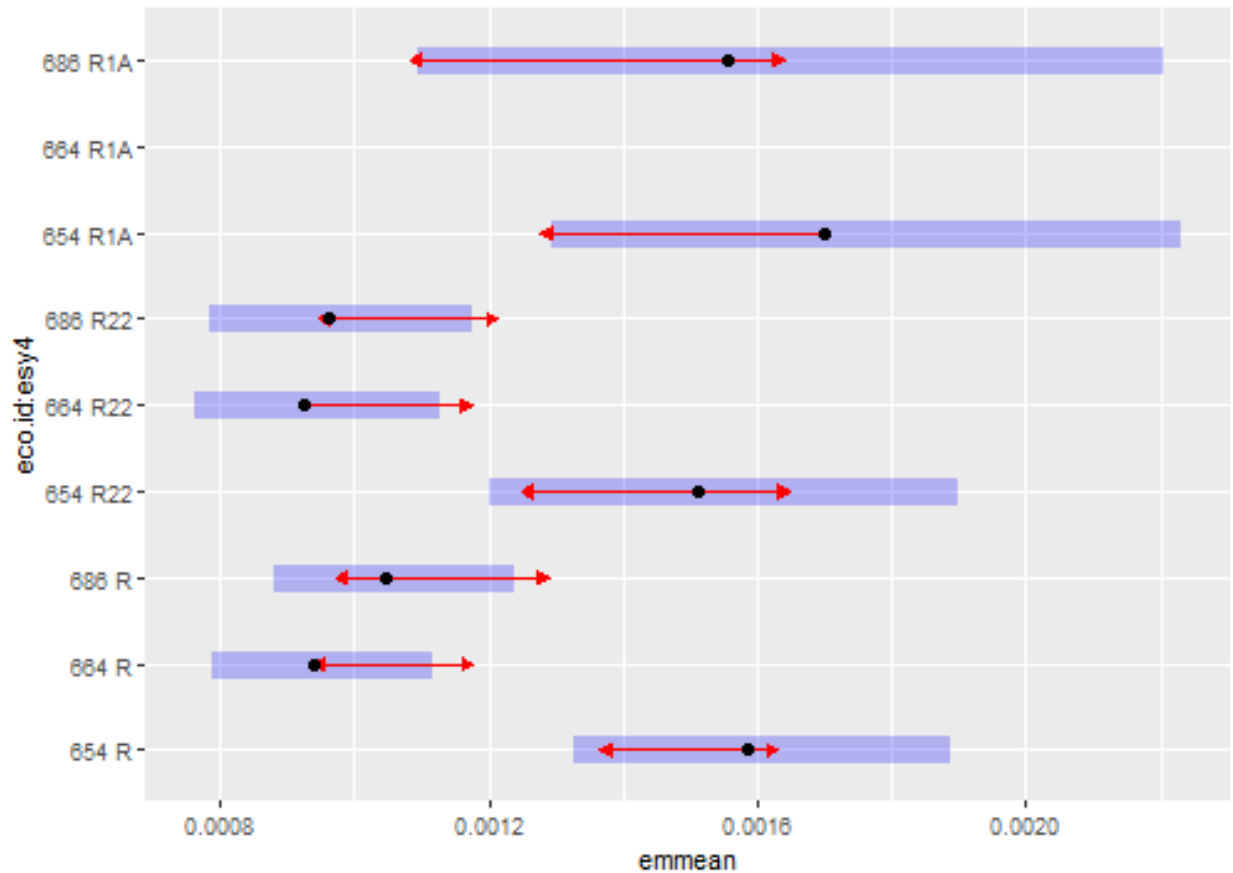
Habitat type x Region

```
## $emmeans
## eco.id esy4 response      SE   df lower.CL upper.CL
## 654    R    0.001583 1.41e-04 200 0.001329 0.00189
## 664    R    0.000939 8.30e-05 215 0.000789 0.00112
## 686    R    0.001046 8.99e-05 183 0.000883 0.00124
## 654   R22    0.001510 1.75e-04 374 0.001202 0.00190
## 664   R22    0.000927 9.27e-05 298 0.000761 0.00113
## 686   R22    0.000961 9.86e-05 296 0.000785 0.00118
## 654   R1A    0.001699 2.35e-04 220 0.001294 0.00223
```

```
## 664 R1A nonEst NA NA NA NA
## 686 R1A 0.001554 2.77e-04 452 0.001095 0.00221
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## contrast ratio SE df null t.ratio p.value
## eco.id664 R / eco.id654 R 0.593 0.0699 233 1 -4.431 0.0004
## eco.id686 R / eco.id654 R 0.660 0.0759 247 1 -3.612 0.0087
## eco.id686 R / eco.id664 R 1.114 0.1280 248 1 0.938 0.9820
## eco.id654 R22 / eco.id654 R 0.954 0.0985 565 1 -0.458 0.9998
## eco.id654 R22 / eco.id664 R 1.609 0.2250 385 1 3.394 0.0172
## eco.id654 R22 / eco.id686 R 1.444 0.1980 402 1 2.681 0.1315
## eco.id664 R22 / eco.id654 R 0.585 0.0749 332 1 -4.183 0.0010
## eco.id664 R22 / eco.id664 R 0.987 0.0954 587 1 -0.132 1.0000
## eco.id664 R22 / eco.id686 R 0.887 0.1100 326 1 -0.970 0.9784
## eco.id664 R22 / eco.id654 R22 0.614 0.0835 381 1 -3.589 0.0089
## eco.id686 R22 / eco.id654 R 0.607 0.0783 352 1 -3.874 0.0032
## eco.id686 R22 / eco.id664 R 1.023 0.1310 334 1 0.177 1.0000
## eco.id686 R22 / eco.id686 R 0.919 0.0786 537 1 -0.992 0.9755
## eco.id686 R22 / eco.id654 R22 0.636 0.0861 393 1 -3.343 0.0203
## eco.id686 R22 / eco.id664 R22 1.036 0.1300 324 1 0.284 1.0000
## eco.id654 R1A / eco.id654 R 1.073 0.1540 309 1 0.488 0.9997
## eco.id654 R1A / eco.id664 R 1.809 0.2980 189 1 3.598 0.0096
## eco.id654 R1A / eco.id686 R 1.625 0.2650 199 1 2.977 0.0635
## eco.id654 R1A / eco.id654 R22 1.125 0.1890 325 1 0.701 0.9969
## eco.id654 R1A / eco.id664 R22 1.833 0.3120 237 1 3.558 0.0105
## eco.id654 R1A / eco.id686 R22 1.769 0.3050 237 1 3.310 0.0236
## eco.id664 R1A / eco.id654 R nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id664 R nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id686 R nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id654 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id664 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id686 R22 nonEst NA NA 1 NA NA
## eco.id664 R1A / eco.id654 R1A nonEst NA NA 1 NA NA
## eco.id686 R1A / eco.id654 R 0.982 0.1940 418 1 -0.093 1.0000
## eco.id686 R1A / eco.id664 R 1.656 0.3290 378 1 2.537 0.1832
## eco.id686 R1A / eco.id686 R 1.487 0.2660 490 1 2.213 0.3457
## eco.id686 R1A / eco.id654 R22 1.029 0.2190 436 1 0.136 1.0000
## eco.id686 R1A / eco.id664 R22 1.677 0.3410 420 1 2.544 0.1801
## eco.id686 R1A / eco.id686 R22 1.618 0.3060 515 1 2.546 0.1789
## eco.id686 R1A / eco.id654 R1A 0.915 0.1700 431 1 -0.477 0.9998
## eco.id686 R1A / eco.id664 R1A nonEst NA NA 1 NA NA
##
## Results are averaged over the levels of: site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 8 estimates
## Tests are performed on the log scale
plot(emm, comparison = TRUE)

## Warning: Removed 1 row containing missing values or values outside the scale range
```

```
## (`geom_point()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



```
# (emm <- emmeans(
#   m_2,
#   reupairwise ~ site.type / esy4,
#   type = "response"
# ))
# plot(emm, comparison = TRUE)
```

Habitat type x Site type

Session info

```
## R version 4.5.0 (2025-04-11 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
## LAPACK version 3.12.1
```

```

##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMA_0.4.7    patchwork_1.3.0 ggbeeswarm_0.7.2
## [5] lubridate_1.9.4 forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4
## [9] purrr_1.0.4     readr_2.1.5     tidyr_1.3.1      tibble_3.2.1
## [13] ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6        magrittr_2.0.3     multcomp_1.4-28
## [7] compiler_4.5.0     mgcv_1.9-1         vctr_0.6.5
## [10] pkgconfig_2.0.3    crayon_1.5.3       fastmap_1.2.0
## [13] backports_1.5.0    labeling_0.4.3     utf8_1.2.5
## [16] ggstance_0.3.7     promises_1.3.2     rmarkdown_2.29
## [19] tzdb_0.5.0         nloptr_2.2.1       bit_4.6.0
## [22] xfun_0.52          later_1.4.2        broom_1.0.8
## [25] parallel_4.5.0     R6_2.6.1           gap.datasets_0.0.6
## [28] stringi_1.8.7      qgam_2.0.0          RColorBrewer_1.1-3
## [31] car_3.1-3          boot_1.3-31        estimability_1.5.1
## [34] Rcpp_1.0.14        iterators_1.0.14    knitr_1.50
## [37] zoo_1.8-14         parameters_0.25.0   httpuv_1.6.16
## [40] Matrix_1.7-3       splines_4.5.0       timechange_0.3.0
## [43] tidyselect_1.2.1   rstudioapi_0.17.1  abind_1.4-8
## [46] yaml_2.3.10        MuMIn_1.48.11       doParallel_1.0.17
## [49] codetools_0.2-20   lattice_0.22-6     plyr_1.8.9
## [52] bayestestR_0.15.3  shiny_1.10.0        withr_3.0.2
## [55] evaluate_1.0.3     marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2      gap_1.6             carData_3.0-5
## [61] foreach_1.5.2      stats4_4.5.0        reformulas_0.4.1
## [64] insight_1.2.0       generics_0.1.4      vroom_1.6.5
## [67] rprojroot_2.0.4    hms_1.1.3           scales_1.4.0
## [70] minqa_1.2.8        xtable_1.8-4        glue_1.8.0
## [73] tools_4.5.0        data.table_1.17.2   lme4_1.1-37
## [76] mvtnorm_1.3-3      grid_4.5.0          rbibutils_2.3
## [79] datawizard_1.1.0   nlme_3.1-168        Rmisc_1.5.1
## [82] performance_0.13.0 beeswarm_0.4.0      vipor_0.4.7
## [85] Formula_1.2-5      cli_3.6.5           gtable_0.3.6
## [88] digest_0.6.37      pbkrtest_0.5.4      TH.data_1.1-3
## [91] farver_2.1.2       htmltools_0.5.8.1   lifecycle_1.0.4
## [94] mime_0.13          bit64_4.6.0-1       dotwhisker_0.8.4
## [97] MASS_7.3-65

```